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MILITARY SPECIFICATION

PROJECTILE, 152MM, TP-T, M411A1
METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This Military Specification covers Projectile, 152MM, TP-T, M411A1, Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this Military Specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-C-60539	- Coatings, Anodic, Conventional, for Aluminum and Aluminum Alloys
MIL-R-11073	- Rotating Band Blanks, Sintered Iron

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-171	- Finishing of Metal and Wood Surfaces

FSC 1320

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2.1 (Cont'd)

MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

F10551938	- Projectile, 152MM, TP-T, M411A1, Metal Parts Assembly
F8837835	- Pallet for Fixed and Semifixed Projectiles
LI-10551938	- Inspection Equipment List for Projectile, 152MM, TP-T, M411A1, Metal Parts Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1	- Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, New York 10018).

ASTM E-8	- Tension Testing of Metallic Materials
ASTM E-18	- Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specification and the following:

3.2 Heat treatment or stress relief.-The heat treatment or stress relief applied shall be uniform throughout each lot of components and shall be so conducted as to produce in any component the mechanical properties specified on the applicable drawings.

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3.2.1 Hot-forged heat-treat method of manufacture (if applicable).- Projectile components manufactured by this method shall be subjected to heat treatment to attain required mechanical properties. The heat treatment of ferrous forgings shall consist of heating above the critical temperature, not to exceed 1600 degrees Fahrenheit (F), followed by an oil quench. After quenching, the piece shall be tempered without delay at a temperature above 700 degrees F., suitable to produce required mechanical properties. After removal from tempering furnace, the projectile components shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700 degrees F., after which cooling may be accelerated by water spray or similar means.

3.2.2 Cold shipping method of manufacture (if applicable).-Cold shaping of steel components shall be so accomplished as to produce in the projectile component the mechanical properties specified. The projectile component shall be stress relieved at a temperature of 700 degrees F., minimum, following the final operation during which time plastic flow takes place. The minimum soak time at or above minimum temperature shall be established which will produce mechanical properties specified by the applicable drawing.

3.3 Rotating band (sintered iron).

3.3.1 Band seat.- Prior to application of retating band blank, the band seat of the body shall be clean and free of oil, grease, dirt, rust and other foreign matter. Shot or sandblasting shall not be permitted for cleaning the band seat after scoring.

3.4 Proving ground functional tests.-The assembled projectile metal parts shall reveal no evidence of loss, breakup, or loosening of any component in the gun bore or in flight when tested at excess and service pressure.

3.5 Metal defects.

3.5.1 Components.-All components, including the cavity, shall be free from cracks, splits, cold shuts, inclusions, pipe, porosity, bursts, folds, seams and other metal defects.

3.5.2 Component cavity.-All component cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

3.5.3 Visual standards.-Visual standards for metal defects will be established for each contract, whenever requested by procuring activity.

3.5.4 Burr.-No component shall have a burr which might interfere with the assembly or function of the round or which might be injurous to personnel handling the item.

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3.6 Protective coating.-Protective coating shall be in accordance with the applicable drawings and the specification referenced thereon. Where protective coating is required by component details, the requirements will apply to all indicated surfaces regardless of contour configuration.

3.7 Painting.-Painting shall comply with the requirements of the applicable drawings. All paint shall be dry to the touch before testing and packing for shipment.

3.8 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawings.

3.8.2 Burr.-No component shall have a burr which might interfere with the assembly or function of the item or which might be injurous to personnel handling the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified on the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define terms used.

4.1.2 Inspection.-Inspection shall be in accordance with Specification MIL-A-2550, except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of forty-five projectile metal parts and five body and band assemblies as follows:

35 metal parts assemblies complete, including paint
5 metal parts assemblies complete, except paint
5 sets of component parts complete, including protective coating except those components requiring protective finish in accordance with MIL-STD-171, finsih no. 5.1.1 shall be furnished without protective coating.
5 body and band assemblies completely machined, less protective coating

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4.2.1 (Cont'd)

Thirty each of the above metal parts assemblies complete including paint shall be forwarded to a designated proving ground for funtional tests. The sample shall be manufactured in the same manner using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply. A first article sample or portion thereof as directed by the procuring agency shall also be submitted whenever a change occurs in the manufacturing process, material used, drawings, specification or source of supply, such as to significantly affect product uniformity as directed by the responsible technical agency.

4.2.1.2 Initial production failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in same disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.2 Forging lots.-Forging lot shall contain not more than one heat of steel.

4.3.1.3 Bar stock lots.-Bar stock lots shall contain not more than one heat of steel.

4.3.1.4 Cold shaping lots.-Cold shaping lots shall contain not more than one heat of steel.

4.3.1.5 Heat treatment lots.-Heat treatment lots shall consist of components manufactured from forgings in accordance with 4.3.1.2, or bar stock in accordance with 4.3.1.3 that have been heat treated in the same furnace at the same temperature in one unchanged process. After final heat treatment, components shall remain grouped for identity purposes until such time as the results of the tension test hereinafter prescribed indicate compliance with mechanical properties of the applicable drawing.

4.3.1.6 Stress relieved lots.-Stress relieved lots shall consist of components manufactured from cold shapings in accordance with 4.3.1.4 that have been stress relieved in the same furnace at the same temperature in one unchanged process. After stress relief, the cold shapings shall remain grouped for identity purposes until such time as the tension test hereinafter prescribed indicates compliance with mechanical properties requirements of the applicable component drawing.

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4.3.1.7 Projectile lots.-When it has been determined that bodies of a heat treated or stress relieved lot have complied with the mechanical properties requirement of the applicable drawing, may be grouped for further processing.

4.3.1.8 Shipping lots.-Shipping lots shall consist of projectile assemblies produced by one supplier, under one contract, in one unchanged process, in accordance with the same drawings, revisions, under conditions of a continuous production, free of interruptions other than those due to the end of shift day, or work week. Shipping lots shall be assigned a lot number in accordance with MIL-STD-1168. The shipping lot thus constituted shall be the basis for selection of samples for the proving ground functional test. Lot size shall be determined by the procuring contracting officer.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ANSI B46.1 shall be used as a basis for roughness determination. Four power (4X) magnification may be used in performing visual examination.

4.3.2.1 Tool control dimensions.-

a. Dimensions marked tool control shall be gaged at the beginning or end of each working day (24 hrs), during First Article Inspection and for the first component after each tooling change, adjustment or repair. Where destruction of the component is necessary to gage such dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production.

b. When the contractor has satisfied (a) above, visual inspection of tool control dimensions will be permitted. Visual inspection shall be performed every two (2) hours for each machine in operation. If any tool control dimension(s) is in doubt during visual inspection, the dimension(s) shall be immediately gaged (see "a" above) to determine acceptability. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

4.3.2.2 Windshield, Drawing D10551941, covering a detail of Drawing F10551938.

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Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
101.	Straightness of ogive	Gage
102.	Pitch diameter of body thread, min	Gage
103.	Major diameter of body thread, min	Gage
104.	Diameter, outside	Gage
105.	Diameter of thread undercut, min. (if applicable)	Gage
106.	Length from rear face to shoulder forward of thread	Gage
107.	Length from shoulder to rear basic diameter	Gage
108.	Length from shoulder to forward basic diameter	Gage
109.	Length of effective thread, min. (if applicable)	Gage
110.	Distance from rear end of thread to shoulder, max. (if applicable)	Gage
111.	Wall thickness, forward end, drill point to nose, min. (if applicable)	Gage
112.	Thread(s) damaged or not full	Visual
113.	Metal defective	Visual
MINOR		
201.	Inside diameter, rear end	Gage
202.	Length from shoulder to front end	Gage
203.	Wall thickness (side wall)	Gage
204.	Wall thickness (front end)	Gage
205.	Radii or chamfers missing or incorrect	Visual (See 6.1)
206.	Surface finish improper	Visual (See 6.1)
207.	Burr	Visual
208.	Foreign matter	Visual

4.3.2.3 Body, Drawing F10551940, covering a detail of Drawing F10551938.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
101.	Runout of rotating band seat with bourrelets	Gage
102.	Diameter of rotating band seat	Gage
103.	Depth of scores in band seat ribs	Visual (See 6.1)
104.	Metal defective	Visual

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4.3.2.3 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Location of rotating band seat	Gage
202. Width of rotating band seat	Gage
203. Evidence of shot or sand blasting of band seat after scoring	Visual
204. Burr	Visual
205. Foreign matter	Visual

4.3.2.4 Body and Band Assembly, Drawing D10551966, covering a detail of Drawing F10551938.

CRITICAL	
None defined	
MAJOR	
101. Runout of body with bourrelets	Gage
102. Runout of inside forward cavity with bourrelets	Gage
103. Runout of rear internal contour with bourrelets	Gage
104. Runout of tracer cavity with bourrelets	Gage
105. Runout of rotating band with bourrelets	Gage
106. True position of tracer cavity with minor diameter of tracer thread	Gage
107. True position of minor diameter at case locking ring thread with minor diameter of tracer thread	Gage
108. Diameter of windshield thread relief, max. (if applicable)	Gage
109. Pitch diameter of windshield thread, max.	Gage
110. Minor diameter of windshield thread, max.	Gage
111. Diameter, outside chamfer, forward end	Gage
112. Diameter of rotating band	Gage
113. Diameter of bourrelets, min	Gage
114. Diameter of tracer cavity	Gage
115. Length from rear face to rear basic diameter in rear cavity	Gage
116. Length from rear face to forward basic diameter in rear cavity	Gage
117. Length from rear face to basic diameter on rotating band	Gage
118. Depth of tracer cavity	Gage
119. Length of effective windshield thread, min.	Gage
120. Length from rear face to bottom of forward cavity	Gage
121. Length from shoulder at end of case locking ring thread to bottom of rear cavity	Gage

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4.3.2.4 (Cont'd)

	Defect	Method of Inspection
122.	Length from forward face to start of windshield thread relief, min. (if applicable)	Gage
123.	Length from rear surface of hub to start of tracer thread relief, min. (if applicable)	Gage
124.	Location of rotating band	Gage
125.	Width of rotating band	Gage
126.	Depth of rear cavity	Gage
127.	Depth of counterbore, forward end	Gage
128.	Thread(s) damaged or not full	Visual
129.	Metal defective	Visual

MINOR

201.	Diameter of body	Gage
202.	Diameter of boss, rear end	Gage
203.	Diameter of forward cavity	Gage
204.	Diameter of counterbore	Gage
205.	Length, total (forward face to end of rear hub)	Gage
206.	Length, total (forward face to end of rear skirt)	Gage
207.	Length from front face to rear of forward bourrelet	Gage
208.	Length from front face to front of rear bourrelet	Gage
209.	Length of windshield cavity	Gage
210.	Depth of plug and disc cavity, max.	Gage
211.	Radii or chamfers missing or incorrect	Visual (See 6.1)
212.	Surface finish improper	Visual (See 6.1)
213.	Burr	Visual
214.	Foreign matter	Visual

4.3.2.5 Plug and disc assembly, Drawing B8595462, covering a detail of Drawing F10551938.

CRITICAL

None defined

MAJOR

101.	True position of inside diameter with major diameter of thread	Gage
102.	Pitch diameter of thread, min.	Gage
103.	Major diameter of thread, min.	Gage
104.	Thickness of disc	Gage
105.	Diameter of thread chamfer	Gage
106.	Width of threaded diameter	Gage
107.	Thread(s) damaged or not full	Visual
108.	Improper assembly	Visual
109.	Metal defective	Visual

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Major diameter of thread, max.	Gage
203. Diameter of through hole	Gage
204. Protective coating inadequate or damaged	Visual
205. Presence and location of wrench holes	Visual (See 6.1)
206. Surface finish improper	Visual (See 6.1)
207. Burr	Visual
208. Foreign matter	Visual

4.3.2.6 Ring, Case Locking, Drawing C10534386, covering a detail of Drawing F10551938.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Thread(s) damaged or not full	Visual
104. Metal defective	Visual
MINOR	
201. Pitch diameter of thread, min.	Gage
202. Minor diameter of thread, min.	Gage
203. Diameter outside	Gage
204. Depth of flat	Gage
205. Location of flat	Gage
206. Width of ring	Gage
207. Width across flats	Gage
208. Radii or chamfers missing or incorrect	Visual (See 6.1)
209. Surface finish improper	Visual (See 6.1)
210. Protective coating inadequate or damaged	Visual
211. Burr	Visual
212. Foreign matter	Visual

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4.3.2.7 Metal Parts Assembly, Drawing F10551938 (prior to paint).

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Runout of front end of windshield with bourrelets	Gage
102. Runout of rear end of windshield with bourrelets	Gage
103. Length, total max.	Gage
104. Insecure assembly (windshield to body)	Torque Wrench (See 4.3.3.5)
105. Presence of gap between body and windshield	Gage
106. Windshield diameter flush or smaller than bourrelet diameter	Gage
MINOR	
201. Marking missing, incorrect or illegible	Visual
202. Burr	Visual
203. Foreign matter	Visual

4.3.2.8 Metal Parts Assembly, Drawing F10551938 (after paint).

CRITICAL	
None defined	
MAJOR	
101. Weight	Approved Scale
102. Pitch diameter of tracer thread, max.	Gage
103. Minor diameter of tracer thread, max.	Gage
104. Pitch diameter of case locking ring thread, min.	Gage
105. Major diameter of case locking ring thread, min.	Gage
106. Diameter of bourrelets, max.	Gage
107. Length of effective tracer thread, min.	Gage
108. Length of effective case locking ring thread, min.	Gage
109. Thread(s) damaged or not full	Visual
110. Rotating band damaged	Visual

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Pitch diameter of tracer thread, min.	Gage
202. Minor diameter of tracer thread, min.	Gage
203. Pitch diameter of case locking ring thread, max.	Gage
204. Major diameter of case locking ring thread, max.	Gage
205. Paint not dry	Tactile
206. Paint on thread(s)	Visual
207. Protective coating inadequate or damaged	Visual
208. Marking missing, incorrect or illegible	Visual (See 6.1)
209. Burr	Visual
210. Foreign matter	Visual

4.3.2.9 Pallet, Drawing F8837835.

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Pallet damaged (unserviceable)	Visual
202. Sleeve(s) missing	Visual
203. Improperly packed	Visual
204. Strapping missing, loose or broken	Visual/Manual
205. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.-The following test shall be performed in accordance with provisions of 4.4.

4.3.3.1 Mechanical properties.

4.3.3.1.1 Hardness.

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4.3.3.1.1.1 Heat treatment lots.--In order to determine uniformity of heat treatment, each component in the heat treatment lot (as defined by 4.3.1.5) shall be subjected to a hardness test.

4.3.3.1.1.2 Stress relieved lots.--From each stress relieved lot (as defined by 4.3.1.6) the first twenty-five components processed through the stress relief furnace and five from each one hour production thereafter for the balance of the stress relieved lot shall be hardness tested to assess uniformity of process. Records of hardness shall be maintained.

4.3.3.1.2 Tension test.

4.3.3.1.2.1 Heat treatment lots.--Two components from each heat treatment lot (one of the hardest and one of the softest as determined by 4.3.3.1.1.1) shall be subjected to this test to determine compliance with the mechanical properties requirements of the applicable drawing. Failure of one or more tensile test specimens to comply with the yield strength or elongation requirement shall be cause for rejection of the lot with the exception that elongation results obtained from tangentially transverse specimens in the base web area of the body shall be recorded for information only. At the option of the supplier a production control sample may be utilized to establish a hardness range for each heat treatment lot. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the supplier. Components which fail outside of the established range may be reheat treated to fall within the range or new mechanical properties may be determined from newly selected components which will represent the true extremes of hardness in the heat treatment lot.

4.3.3.1.2.2 Stress relieved lots.--Two components from each stress relieved lot shall be subjected to this test to determine compliance with the mechanical properties of the applicable drawing. Failure of one or more tensile test specimens to comply with the yield strength or elongation requirements shall be cause for rejection of the lot with the exception that elongation results obtained from tangentially transverse specimens in the base web area of the body shall be recorded for information only.

4.3.3.1.2.3 Bar stock lot.--With the approval of the procuring activity when components are manufactured from bar stock in such a manner that the mechanical properties are not altered, three bars, from each bar stock lot may be tested in lieu of the component. Failure of any tensile test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the bar stock lot.

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4.3.3.2 Aluminum components.-The test procedures, sample sizes and acceptance criteria shall be as specified in the applicable material specification. Failure to comply with the requirements of the applicable drawing and specification shall be cause for rejection of the material represented by the sample.

4.3.3.3 Rotating band.

4.3.3.3.1 Band seating.-The body and band assemblies from each lot shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Failure of either assembly to comply shall be cause for rejection of the lot.

4.3.3.3.2 Density.-The sample provided by 4.3.3.3.1 after examination for seating shall be subjected to this test to determine compliance with requirements of the applicable drawing. Failure of any specimen to comply shall be cause for rejection of the lot.

4.3.3.4 Cleaning and application of protective coating.-The sample size shall be as specified in the applicable specification with the exception that the sample size for the corrosive resistance test for those components requiring protective finish in accordance with specification MIL-C-60539, Type II, CLASS I shall be selected in accordance with MIL-STD-105, Inspection Level S-2. The acceptance criteria shall be as specified in the applicable specification. Failure to comply with the requirements of the applicable drawings and specification shall be cause for taking action prescribed by the applicable specification.

4.3.3.5 Torque test.-Metal parts assemblies shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Failure of the sample to comply with the criteria of the sampling plan as evidenced by any preceptible movement shall be cause for rejection of the lot.

4.3.4 Proving ground functional tests.

4.3.4.1 First article sample.

4.3.4.1.1 Initial production sample.-The quantities listed in Table I shall be ballistically tested in the phases indicated.

TABLE I

Test Phase	Quantity
Excess Pressure	15
Service Pressure	15

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4.3.4.2 Production lot sample.-The quantities listed in Table II shall be ballistically tested in the phases indicated.

Table II

Test Phase	Quantity
Excess Pressure	10
*Service Pressure	10

*After three (3) consecutive production lots have been ballistically tested and accepted, ballistic testing for this phase may be discontinued. However, implementation of ballistic testing will be instituted whenever directed by the procuring activity.

4.3.4.3 Excess pressure.-Samples provided in accordance with Tables I and II shall be subjected to this test. Failure of any projectile assembly to comply with 3.4 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.4.4 Service pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.4 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.5 Inspection equipment.-Index of inspection equipment list LI-10551938 identifies the inspection equipment required to perform the examinations and tests prescribed herein.

4.4 Test methods and procedures.

4.4.1 Mechanical properties.

4.4.1.1 Hardness.-After final heat treatment or stress relief and before final machining, a Rockwell Hardness Test shall be performed on the body between the bourrelets on the body diameter and on the rear outside surface windshield. The test shall be made using the procedure for performing this test as prescribed by ASTM Method E-18.

4.4.1.2 Tension test.

4.4.1.2.1 Yield strength and elongation (heat treatment and stress relieved lots).-From the components selected in accordance with 4.3.3.1.2.1 and 4.3.3.1.2.2, tensile test specimens shall be selected as follows: From each body, one tensile test specimen shall be taken longitudinally from midpoint of the body sidewall and one tensile test specimen shall be taken from the body in a tangentially transverse direction from the central portion

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4.4.1.2.1 (Cont'd)

of the base web area. The centerline of this specimen shall be located two (2) inches plus ½ inch from the axis of the body. The body specimens shall conform to the largest obtainable standard "R" type specimen. From each windshield, two full length test specimens shall be taken longitudinally 180 degrees apart. They shall conform to the largest obtainable standard "T" type specimen. All specimen types shall conform to ASTM Method E-8. Yield strength shall be determined by the offset method utilizing the procedures of ASTM Method E-8. At the option of the supplier, a standard "R" type specimen or "F" type specimen, as described by Federal Test ASTM Method E-8, may be utilized for making yield strength and elongation determinations on the heat treated or stress relieved windshield prior to final machining, provided the largest obtainable specimens are no smaller than a standard R3 specimen or F2 specimen. The largest obtainable standard "F" type specimen will be used by the Government for initial production lot verification and in any subsequent Government testing of the windshield as may be necessary.

4.4.1.2.2 Yield strength and elongation (bar stock lots).-From one end of each bar selected in accordance with 4.3.3.1.2.3, one longitudinal tensile test specimen shall be removed. It shall be located at mid-radius of the bar. The specimens shall conform to the largest obtainable standard "R" type specimen as described in ASTM Methods E-8. Yield strength shall be determined by the offset method utilizing the procedures of ASTM Methods E-8.

4.4.1.3 Aluminum components.-Procedures for performing the mechanical properties tests shall be in accordance with the criteria prescribed by the applicable material specification.

4.2.2 Rotating band.

4.4.2.1 Band seating.-The body and band assemblies provided by 4.3.3.3 shall be carefully sectionalized with a wet abrasive cutoff wheel, so as to produce 4 segments of 90 degrees each. Visual examination of the exposed surfaces of the rotating band shall then be conducted to determine compliance with the band seating requirements of the applicable drawing.

4.4.2.2 Density.-After examination for band seating as prescribed by 4.4.2.1, two specimens from each body and band assembly spaced approximately 180 degrees apart shall be removed from the rotating band. The specimens shall be the full width and thickness of the band and approximately one inch long. The specimens shall be sawed by hand to the specified size. With the exception of removing the burrs, no further machining shall be performed on the specimens. After weighing the specimens, the wax shall be removed by either one of the two methods described in "a" and "b".

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4.4.2.2 (Cont'd)

a. Place specimens in a dry hydrogen atmosphere furnace and hold at 1400 degrees F. for one hour. Remove specimens and weigh. Return specimens to the furnace and hold at 1400°F. for 30 minutes. This procedure with the half hour heating intervals shall be repeated until the change in weight is less than .05%.

b. A Soxhlet apparatus of suitable size using either tolune or trichloroethylene may be used for extracting the wax. The extraction time shall be a minimum of 18 hours. The specimens shall then be dried for 1½ hours in an oven at 210 degrees F. and then weighed. This procedure with 1½ hour extraction intervals shall be repeated until the change in weight is less than .05%.

Following wax removal and weighing, the specimens shall be impregnated with micro-crysalline wax. This shall be accomplished by immersing the specimen in micro-crystalline wax, held at temperature 180 degrees F. to 200 degrees F. The pressure shall then be reduced by a suitable evacuating method to not more than 2 inches of mercury and held for 30 minutes. The pressure shall then be slowly increased (approximately 30 seconds) to atmospheric pressure after which the specimen shall remain immersed in the molten micro-crystalline wax at 180 degrees F. to 200 degrees F. for approximately one hour.

The density of the specimen shall be calculated as follows:

$$D = \frac{A}{B - C}$$

Where:

D = density in grams per cubic centimeter
A = weight in air of wax free specimen, in grams
B = weight in air of micro-crystalline wax impregnated specimens in grams
C = weight in micro-crystalline wax impregnated specimen immersed in water, in grams

4.4.2.3 Density and wax content of band blank.-The procedure for determining density and wax content shall be as specified in 4.4.2.2 except that from the band blanks selected in accordance with the applicable paragraphs in MIL-R-11073, nine specimens shall be sectioned from each band. They shall consist of three sets spaced approximately 120 degrees apart. The approximate size of each specimen shall be one inch long 1/3 the width of the band x the full thickness of the band (See Figure I). The specimens shall be sawed by hand to the specified size. With the exception of removing burrs, no further machining shall be performed on the specimens.

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4.4.3 Protective coating.-The procedure for testing the cleaning of components and the application of protective coating shall be in accordance with the criteria for testing as specified in the applicable specification.

4.4.4 Torque test.-The assemblies selected in accordance with 4.3.3.5 shall be mounted in a suitable holding device in such a manner that the body component is prevented from turning. The applicable torque shall be applied to the windshield in the direction of the assembly. Testing shall be conducted while thread sealant is wet.

4.4.5 Proving ground functional tests.

4.4.5.1 Excess pressure.-The ballistic test projectiles will be fired in an acceptable standard gun based upon current gun tube condemnation criteria. A propellant charge adjusted to produce 109 to 117 per cent of the upper pressure limit of the propellant will be used. Photographic equipment (smear camera) will be set up near the gun muzzle to record the flight of the projectile for compliance with 3.4. Visual observation will be made to assure projectile impacts in target area. Velocity and chamber pressure will be recorded.

4.4.5.2 Service pressure.-The ballistic test projectiles will be fired in an acceptable standard gun based upon current gun tube condemnation criteria. The propellant shall be of a standard service charge. The test rounds will be fired alternately with projectiles from a reference lot for comparison. All rounds will have velocities recorded (information see 6.2) and will be fired at a vertical target placed 2000 meters from the gun muzzle. Accuracy in terms of maximum spread in inches, both horizontal and vertical and target dispersion in standard deviation in mils shall be recorded for all rounds fired (information see 6.2). Photographic equipment will be set up near the gun muzzle in such a manner as to record the flight of the projectile prior to its impact with the target. Photographs will be examined for compliance with requirements of 3.4.

5. PREPARATION FOR DELIVERY

5.1 Preparation and packing.

5.2 Packing.

5.2.1 Level A.-Not applicable.

5.2.2 Level B.-Not applicable.

5.2.3 Level C - Minimum military pack.-The metal parts for Projectile, 152MM, TP-T, M411A1, shall be placed as follows for shipment to the designated loading facility.

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5.2.3.1 Metal parts assembly.-Part No. 10551938 shall be packed in accordance with Drawing F8837835.

5.2.3.2 Plug and disc assembly.-Part No. 8595462 shall be packed in accordance with MIL-STD-1169. Pack 2500 plug and disc assemblies arranged in a 10 x 10 pattern, 25 to a cell, using interlocking separators.

5.2.3.3 Ring, case locking.-Part No. 10534384 shall be packed in accordance with MIL-STD 1169. Pack 400 rings, case locking arranged in a 4 x 5 pattern, two layers, 10 to a cell, using interlocking separators.

5.3 Shipping.-When components parts from more than one lot are shipped as a carload or truckload, each lot must be kept separate and the division between lots clearly indicated.

5.4 Marking.-The pallet shall be marked in accordance with Drawing F8837835. Cartons shall be marked in accordance with MIL-STD-1169.

5.5 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual examination qualifications.-When compliance with the applicable requirement is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Velocity - Accuracy.-Proving ground tests for velocity-accuracy are not acceptance tests, but the results will be carefully analyzed by DARCOM to determine quality of ammunition produced. Evidence of excessive variation in any case of the Proving Ground tests will result in immediate investigation by DARCOM.

6.3 Visual standards.-Visual standards to be established at contractor's plant in compliance with 3.5.3.

6.4 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

Custodian:

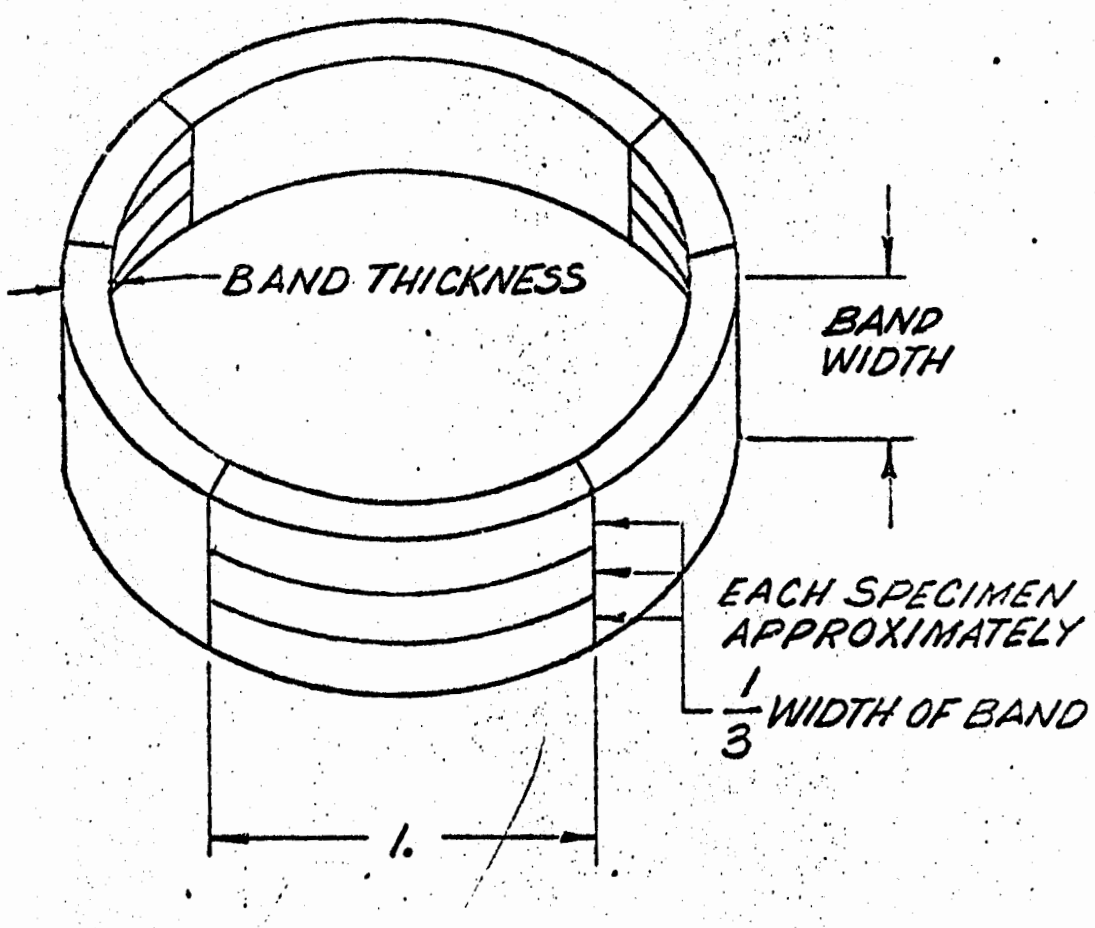
Army - MU

Preparing activity:

Army - MU

Project No. 1320-A328

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THREE SETS OF SPECIMENS AS INDICATED, SHALL BE SECTIONED FROM EACH BAND TO BE TESTED, SPACED APPROXIMATELY 120° APART.

FIG. 1

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